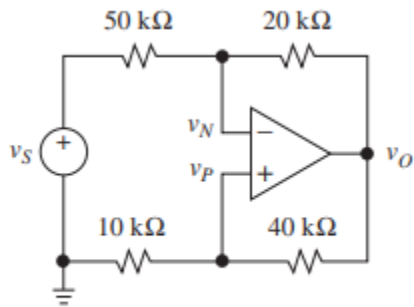


P1) Book Problem 5.4

Investigate the effect of I_B and I_{OS} on the performance of the circuit of Fig. P1.17 if $I_B = 100 \text{ nA}$ flowing out of the op amp's input pins, and $I_{OS} = 10 \text{ nA}$.

**FIGURE P1.17**

if $I_n = 100 \text{ nA}$, $I_p = 0$, $V_s = 0$

$$V_p = 10\text{k}/(10\text{k}+40\text{k})V_o = 0.2 V_o$$

$$V_n = 0.2 V_o$$

$$I_n = (V_o - V_n)/20\text{k} + (0 - V_n)/50\text{k}$$

$$I_n = V_o/20\text{k} - V_n/20\text{k} - V_n/50\text{k}$$

$$I_n = V_o/20\text{k} - 5V_n/100\text{k} - 2V_n/100\text{k}$$

$$I_n = V_o/20\text{k} - 7V_n/100\text{k}$$

$$I_n = V_o/20\text{k} - 7(0.2 V_o)/100\text{k}$$

$$I_n = 0.000,036 V_o$$

$$V_o = 27,777 I_n$$

if $I_n = 0$, $I_p = 100 \text{ nA}$, $V_s = 0$

$$V_n = 50\text{k}/(50\text{k}+20\text{k})V_o = 5/7 V_o$$

$$V_p = 5/7 V_o$$

$$I_p = (V_o - V_p)/40\text{k} + (0 - V_p)/10\text{k}$$

$$I_p = V_o/40\text{k} - V_p/40\text{k} - V_p/10\text{k}$$

$$I_p = V_o/40\text{k} - V_p/40\text{k} - 4V_p/40\text{k}$$

$$I_p = V_o/40\text{k} - 5V_p/40\text{k}$$

$$I_p = V_o/40\text{k} - 5(5/7 V_o)/40\text{k}$$

$$I_p = V_o/40\text{k} - 25/7 V_o/40\text{k}$$

$$I_p = -18/7 V_o/40\text{k}$$

$$I_p = -18/7 V_o/40\text{k}$$

$$I_p = -0.0000642857142857 V_o$$

$$V_o = -15,555 I_p$$

$$V_o = 27,777 I_n - 15,555 I_p$$

$$V_o = 27,777 (I_B + I_{OS}/2) - 15,555 (I_B - I_{OS}/2)$$

$$V_o = 27,777 I_B + 27,777 I_{OS}/2 - 15,555 I_B + 15,555 I_{OS}/2$$

$$V_o = 12,222 I_B + 21,666 I_{OS}$$

$$V_o = -12,222 \cdot 100\text{n} - 21,666 \cdot (+- 10\text{n})$$

$V_o = -0.001222 \pm 0.00021666$

$V_o = -1.222\text{mV} \pm 0.21666\text{mV}$

$V_{oe \text{ min}} = -1.00555\text{mV}$

$V_{oe \text{ max}} = -1.43888\text{mV}$

P2) -6050 Only: Book Problem 5.10

Investigate the effect of I_B and I_{OS} in the current amplifier of Fig. 2.12a. How would you modify the circuit to minimize its dc error?

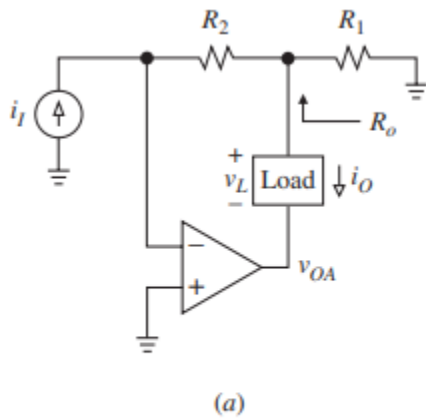


FIGURE 2.12

Current amplifiers: (a) floating-load type,

I_p has no effect on the output

I_n connects directly to the input, so it gets amplified same as the input

$$I_{oe} = -(1 + R_2/R_1)I_n = -(1 + R_2/R_1)(I_B + I_{OS}/2)$$

Add resistor to I_p

$$V_p = -I_p R_p$$

$$V_n = V_p$$

$$0 - V_p = I_o R_1 = I_p R_p$$

$$I_o = I_p R_p / R_1$$

$$I_o = R_p / R_1 I_p - (1 + R_2/R_1) I_n$$

$$I_o = R_p / R_1 (I_B - I_{OS}/2) - (1 + R_2/R_1) (I_B + I_{OS}/2)$$

$$I_o = (R_p/R_1 - 1 - R_2/R_1) I_B - (1 + R_p/R_1 + R_2/R_1) I_{OS}/2$$

for minimal error, minimize I_B factor, $(R_p/R_1 - 1 - R_2/R_1)$

$$R_p/R_1 - R_2/R_1 = 1$$

$$\text{set } R_p = R_1 + R_2$$

$$I_o = ((R_1 + R_2)/R_1 - 1 - R_2/R_1) I_B - (1 + (R_1 + R_2)/R_1 + R_2/R_1) I_{OS}/2$$

$$I_o = - (1 + 1 + R_2/R_1 + R_2/R_1) I_{OS}/2$$

$$I_o = - (1 + R_2/R_1) I_{OS}$$

P3) Book Problem 5.20

Suppose the I-V converter of Fig. 2.2 is implemented with $R = 500\text{ k}$, $R_1 = 1.0\text{ k}$, and $R_2 = 99\text{ k}$, and a FET-input op amp. The op amp has $a = 100\text{ dB}$, $\text{PSRR} = 80\text{ dB}$, and $V_{OS} = 0$, and is powered from dual supplies $\pm V_S = \pm[5 + 1 \sin(2\pi t)]\text{ V}$. (a) Sketch and label the circuit, and following the reasoning in connection with Fig. 5.14, convert it to an equivalent circuit that uses an op amp with $a = \infty$ but with a suitable input offset voltage V_{OS} to account for the finite a and finite PSRR. (b) If we write $v_O = A_{\text{ideal}} i_I + E_O$, what are A_{ideal} and E_O ?

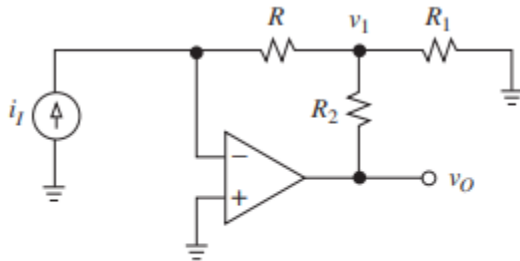


FIGURE 2.2
High-sensitivity I - V converter.

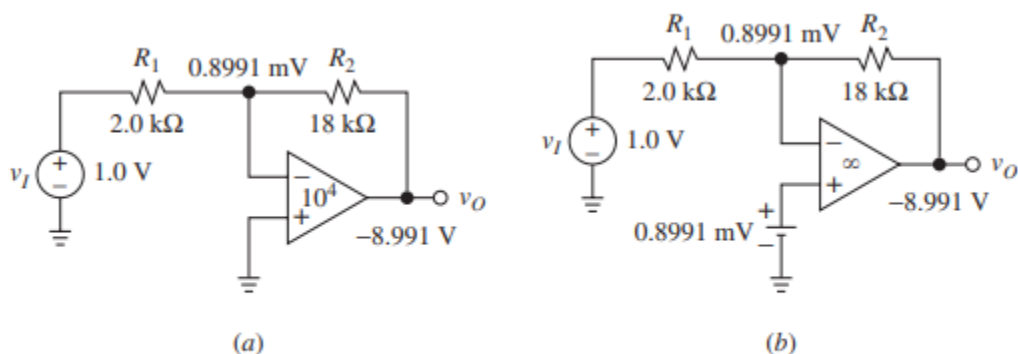
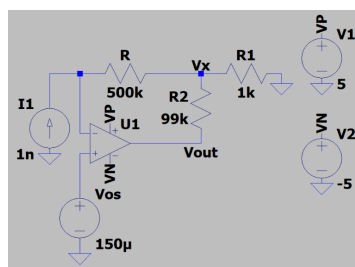
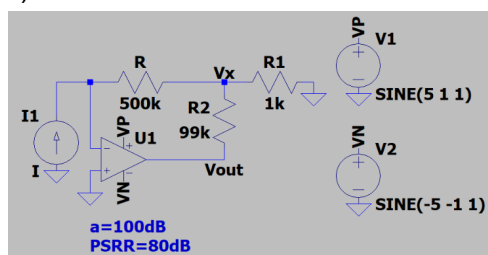


FIGURE 5.14
Using the input offset voltage to model the effect of finite open-loop gain a .

a)



Ideal:

$$I(R) = I_{in}$$

$$V(R) = I_{in} \cdot R$$

$$V_x = -I_{in} \cdot R$$

$$I(R1) = V_x / R1 = -I_{in} \cdot R / R1$$

$$I(R2) = I(R) - I(R1) = I_{in} + I_{in} \cdot R / R1 = I_{in} \cdot (1 + R / R1)$$

$$V(R2) = I_{in} \cdot (1 + R / R1) \cdot R2 = I_{in} \cdot (R2 + R \cdot R2 / R1)$$

$$V_{out} = -I_{in} \cdot R - I_{in} \cdot (R2 + R \cdot R2 / R1) = -I_{in} \cdot (R + (R2 + R \cdot R2 / R1)) = -I_{in} \cdot (R + R2 + R \cdot R2 / R1)$$

$$V_{out} = -I_{in} (500k + 99k + 500k \cdot 99k / 1k) = -I_{in} (599k + 500 \cdot 99k) = -I_{in} (50.099Meg)$$

Add $V_{os} = V_p$

$$V_n = V_p = V_{os}$$

V_x increases by V_{os}

$$I(R1) \text{ decreases from } V_x / R1 \text{ to } (V_x - V_{os}) / R1 = V_x / R1 - V_{os} / R1$$

$$I(R2) \text{ decreases by } V_{os} / R1$$

$$V(R2) \text{ decreases by } R2 \cdot V_{os} / R1$$

$$V_{out} \text{ changes by } V_{os} - R2 \cdot V_{os} / R1 = V_{os} (1 - R2 / R1)$$

$$V_{out} = -I_{in} \cdot (R + R2 + R \cdot R2 / R1) + V_{os} (1 - R2 / R1)$$

$$V_{out} = -I_{in} (50.099Meg) + V_{os} (1 - 99k / 1k) = -I_{in} (50.099Meg) + V_{os} (-98)$$

Add a :

$$a = 100dB = 10^{100/20} = 100k$$

$$V_{osa} \cdot a = V_{outos}$$

$$V_{out} \text{ max} = 5V$$

$$V_{osa} = 5V / 100k = 0.05mV = 50\mu V$$

Add PSRR

$$PSRR = 80dB = 10^{80/20} = 10k$$

$$\Delta V_s = \sin(2\pi t)$$

$$\Delta V_{out} = \sin(2\pi t) / 10k = 100\mu \cdot \sin(2\pi t)$$

$$V_{os} = 150\mu V$$

b)

$$A_{ideal} = 50.099Meg = 50V/\mu A$$

$$E_o = \pm (5mV + 10mV \sin(2\pi t))$$

P4) Book Problem 5.27: This problem presents different contributions for VOS, which are all captured by Equation 5.27. For each part of the question, determine the applicable contributions to VOS; where some may be neglected.

An op amp with $a_{min} = 10^4$ V/V, $V_{OS0}(\max) = 2$ mV, and $CMRR_{dB}(\min) = PSRR_{dB}(\min) = 74$ dB is configured as a voltage follower. (a) Estimate the worst-case departure of v_O from the ideal for $v_I = 0$ V. (b) Repeat with $v_I = 10$ V. (c) Repeat if the supply voltages are lowered from ± 15 V to ± 12 V.

$$V_{OS} = V_{OS0} + TC(V_{OS})\Delta T + \frac{\Delta v_P}{CMRR} + \frac{\Delta V_S}{PSRR} + \frac{\Delta v_O}{a} \quad (5.27)$$

74dB = 5011.87

a)

$V_I = 0$ V

$V_{cm} = 0$ V

$V_{OS0} = 2$ mV

$V_{OS} = V_{OS0} = 2$ mV

$E_o = 2$ mV

b)

$V_I = 10$ V

$V_{cm} = 10$ V

$V_{OS} = V_{OS0} + V_{cm}/CMRR + V_O/a = 2\text{mV} + 10\text{V}/5012 + 10\text{V}/10,000 = 2\text{mV} + 2\text{mV} + 1\text{mV} = 5\text{mV}$

$E_o = 5$ mV

c)

$V_{OS_{new}} = V_{OS_{old}} + \Delta V_S/PSRR = V_{OS_{old}} + 3\text{V}/5012 = V_{OS_{old}} + 0.6\text{mV}$

ca) 2.6mV

cb) 5.6mV